

Multi-Camera 3D Coordinate Capture System for Clinical Scoring

Analysis Report

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1) 3D Scatter Plot Analysis: Raw vs. Calibrated Face Points

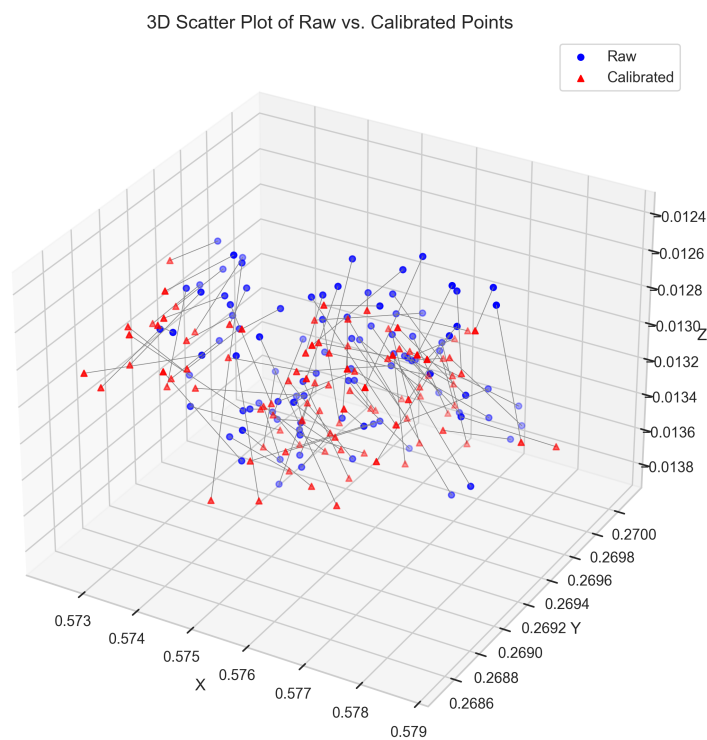


Figure 1.1: 3D Scatter plot showing raw (blue) vs. calibrated (red) face landmark points.

1.1. Detailed Explanation

1.1.1. Axes and Point Representation

- Axes (X, Y, Z):
 - Represent the normalized 3D coordinate space for face landmarks.

- **X-axis:** Horizontal position of the landmark in the image.
- **Y-axis:** Vertical position of the landmark in the image.
- **Z-axis:** Depth information (distance from the camera).
- **Blue Dots (Raw Points):**
 - Original 3D coordinates of face landmarks detected in frames before calibration.
 - They may exhibit distortion due to lens imperfections.
- **Red Dots (Calibrated Points):**
 - Corrected coordinates after applying calibration to remove distortion.
 - Ensure spatial accuracy by accounting for lens distortion.
- **Gray Lines:**
 - Connect each raw (blue) point to its corresponding calibrated (red) point.
 - Their length and direction indicate the magnitude and orientation of the correction.

1.1.2. Step-by-Step Analysis

1. Overall Distribution of Points:

- Clusters of raw (blue) and calibrated (red) points populate the normalized coordinate space.
- Observations indicate that while both sets are generally clustered, the calibrated points are slightly shifted due to correction.

2. Dispersion and Shifts:

- **Short Gray Lines:** Represent minimal shifts, common for landmarks near the image center.
- **Long Gray Lines:** Represent larger shifts, frequently observed at the periphery where lens distortion is stronger.

3. Systematic Offsets:

- Calibrated (red) points appear systematically closer to a central cluster compared to raw (blue) points.
- This suggests an inward correction addressing radial distortion.

1.1.3. Visual Verification and Diagnostics

- The plot visually confirms that calibration reduces distortion by aligning raw points closer to their true positions.
- Anomalies, such as erratic or excessively large shifts, may indicate issues with calibration parameters or insufficient calibration image coverage.

2) Basic coordinates comparison plots

2.1. Plot 1: Euclidean Distance Error (Combined 3D Error)

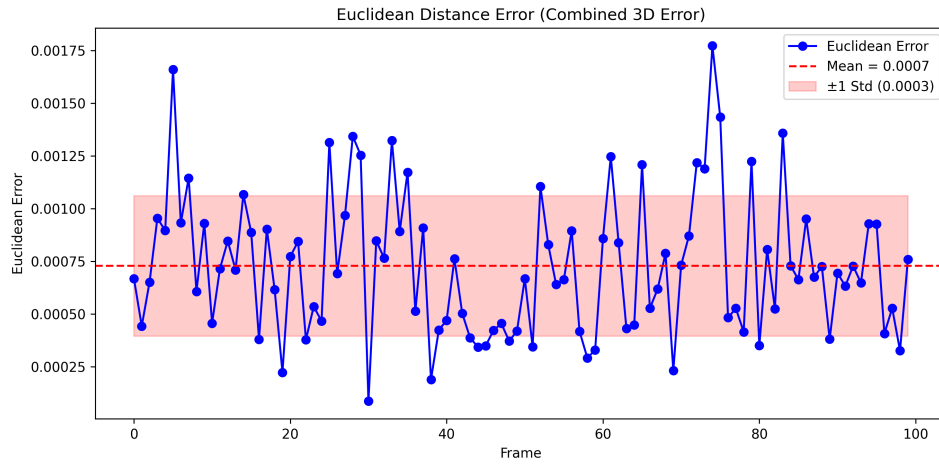


Figure 2.1: Euclidean Distance Error Plot.

2.1.1. Overview

- **Y-axis:** Represents the Euclidean error (the 3D distance between raw and calibrated face landmarks for each frame).
- **X-axis:** Denotes the frame number (time progression).
- **Key Elements:**
 - **Blue Line:** Displays the Euclidean error for each frame.
 - **Red Dashed Line:** Indicates the mean Euclidean error across all frames (0.0007).

- **Shaded Region (Pink):** Represents ± 1 standard deviation (0.0003) around the mean.

2.1.2. Interpretation

1. **Error Magnitude:** The consistently small Euclidean error indicates that calibration introduces only minimal shifts in landmark positions. A mean error of 0.0007 reflects high calibration accuracy.
2. **Error Stability:** The majority of errors fall within the pink shaded region (± 1 standard deviation), though occasional spikes may occur when distortion is more pronounced.
3. **Clinical Relevance:** The small and consistent error values are crucial for reliable landmark detection in clinical scoring systems such as ARAT and UPDRS.

2.2. Plot 2: Per-Axis Error (X, Y, Z)

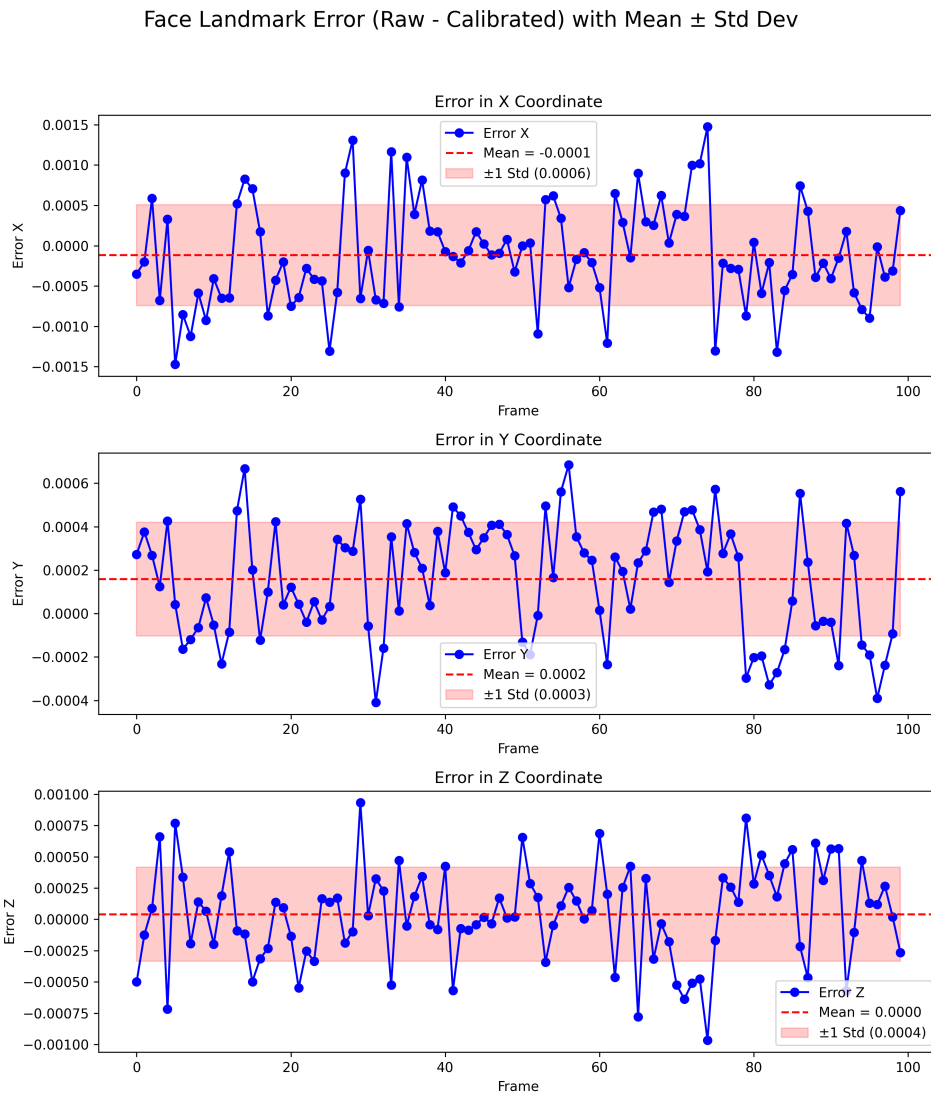


Figure 2.2: Per-Axis Error Plot for X, Y, and Z.

2.2.1. Overview

This plot dissects the error into its three spatial components:

- **Top Graph (X):** Error in the horizontal position.
- **Middle Graph (Y):** Error in the vertical position.
- **Bottom Graph (Z):** Error in the depth (distance from the camera).

Each graph comprises:

- **Blue Line:** Per-frame error for the respective coordinate.
- **Red Dashed Line:** Mean error for that axis.
- **Shaded Region (Pink):** ± 1 standard deviation around the mean.

2.2.2. Interpretation

X-Axis Error: Mean error is approximately -0.0001 with a standard deviation of 0.0006, indicating minimal and stable horizontal corrections.

Y-Axis Error: A slightly positive mean error of 0.0002 with a smaller standard deviation of 0.0003 suggests consistent vertical corrections.

Z-Axis Error: With a mean error nearly zero (0.0000) and a standard deviation of 0.0004, depth corrections are precise.

2.3. Plot 3: Raw vs. Calibrated Coordinates

Face Landmark Coordinates: Raw vs. Calibrated

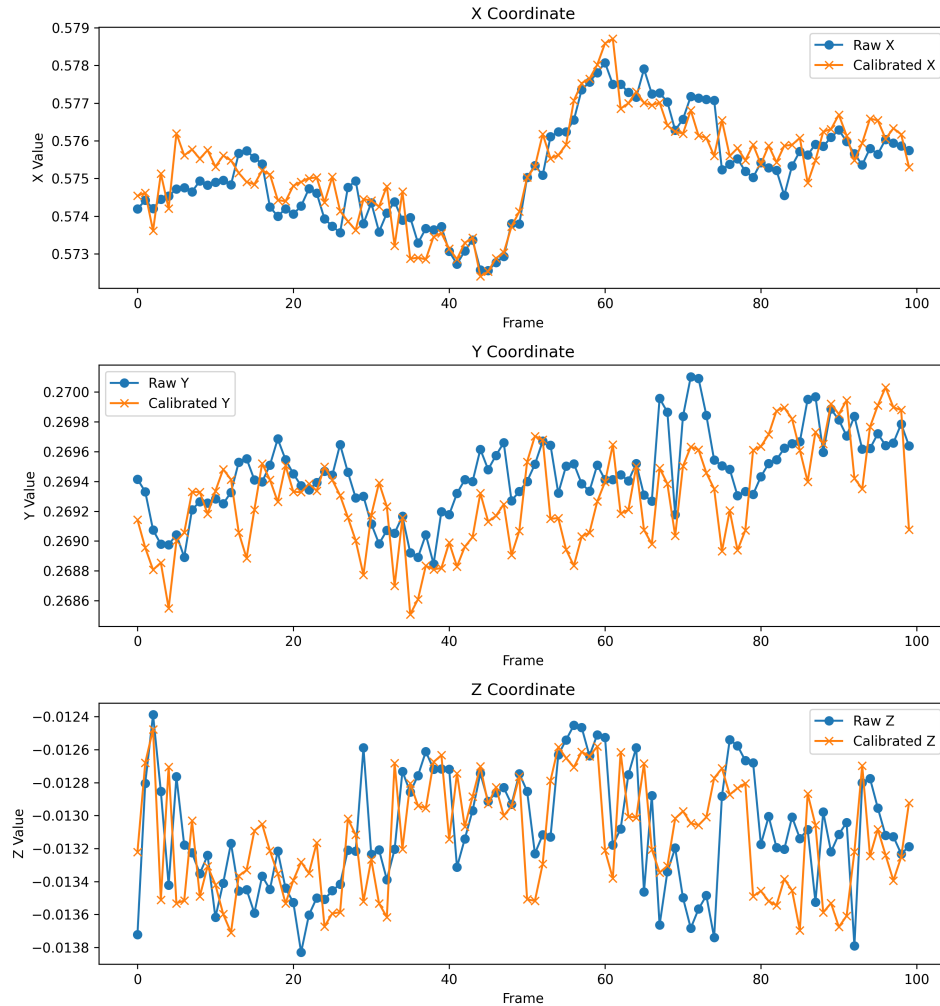


Figure 2.3: Comparison of Raw vs. Calibrated Coordinates for X, Y, and Z.

2.3.1. Overview

This plot compares the raw and calibrated coordinates for each of the three axes:

- **Blue Line:** Represents raw coordinate values over frames.
- **Orange Line:** Represents calibrated coordinate values over frames.

2.3.2. Interpretation

1. **X Coordinate Comparison:** The close overlap indicates that horizontal corrections are minimal.
2. **Y Coordinate Comparison:** Similarly, vertical corrections are slight, with calibrated values showing minor adjustments.
3. **Z Coordinate Comparison:** Depth values align closely, with minimal shifts that reflect effective depth correction.
4. **Clinical Relevance:** The consistency between raw and calibrated data confirms that calibration does not introduce significant artifacts, preserving the integrity of clinical measurements.

2.4. Overall Insights from All Plots

1. Calibration effectively corrects lens distortion while maintaining spatial accuracy across all dimensions.
2. Errors are consistently small and stable across frames, supporting the robustness of the calibration process.
3. The close alignment between raw and calibrated coordinates ensures reliable face landmark detection, which is critical for clinical scoring tasks such as ARAT, UPDRS, and Fugl-Meyer assessments.

3) Bland–Altman Plots for Face Calibration

3.1. Introduction

Bland–Altman plots assess the agreement between two measurement methods—in this case, the raw (uncalibrated) and calibrated face landmark coordinates. In the following, each plot corresponding to the X, Z, and Y coordinates is analyzed in detail.

3.2. Plot 1: Bland–Altman Plot for X Coordinate

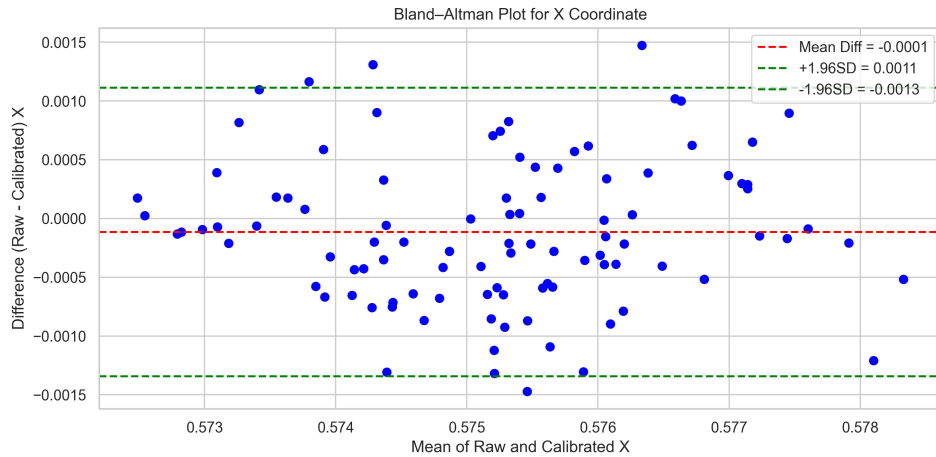


Figure 3.1: Bland–Altman Plot for X Coordinate.

3.2.1. Overview

- **X-axis:** Mean of raw and calibrated X coordinates for each landmark.
- **Y-axis:** Difference between raw and calibrated X coordinates (Raw – Calibrated).
- **Key Elements:**
 - **Red Dashed Line:** Mean difference (-0.0001), representing systematic bias.

- **Green Dashed Lines:** Limits of agreement (± 1.96 SD), ranging from -0.0013 to 0.0011 .

3.2.2. Interpretation

1. **Mean Difference:** The mean difference is nearly zero (-0.0001), indicating negligible systematic bias for the X coordinate.
2. **Limits of Agreement:** Most differences lie within the limits (-0.0013 to 0.0011), showing that calibration adjusts the X coordinate consistently.
3. **Scatter Distribution:** Blue dots are evenly scattered around the mean difference line with slightly larger differences near the image edges.
4. **Clinical Relevance:** The small bias and narrow limits confirm that horizontal positioning is preserved after calibration.

3.3. Plot 2: Bland–Altman Plot for Z Coordinate

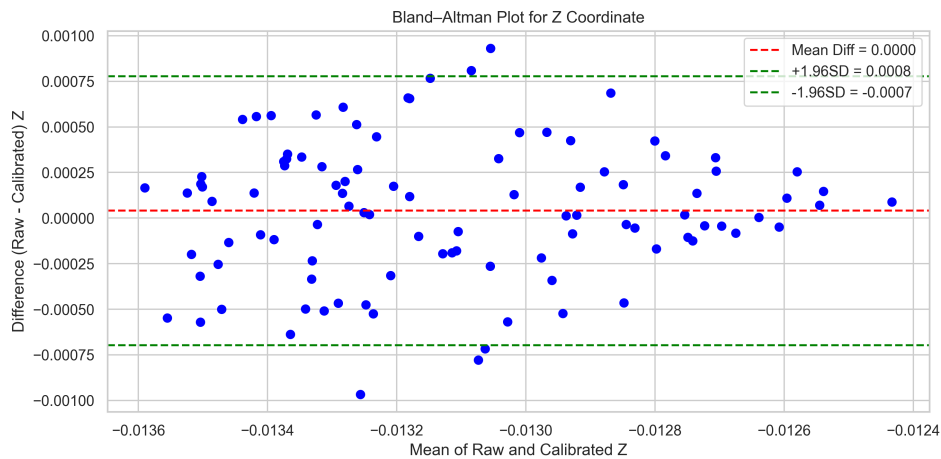


Figure 3.2: Bland–Altman Plot for Z Coordinate.

3.3.1. Overview

- **X-axis:** Mean of raw and calibrated Z coordinates for each landmark.
- **Y-axis:** Difference between raw and calibrated Z coordinates (Raw – Calibrated).

- **Key Elements:**

- **Red Dashed Line:** Mean difference (0.0000), indicating no systematic bias.
- **Green Dashed Lines:** Limits of agreement (± 1.96 SD), ranging from -0.0007 to 0.0008 .

3.3.2. Interpretation

1. **Mean Difference:** A mean difference of 0.0000 shows that no bias is introduced in depth correction.
2. **Limits of Agreement:** The narrow range (-0.0007 to 0.0008) reflects high precision in depth corrections.
3. **Scatter Distribution:** Blue dots are symmetrically distributed around the mean line, with no evident trends.
4. **Clinical Relevance:** Precise depth correction is critical to maintain reliable 3D spatial representation.

3.4. Plot 3: Bland–Altman Plot for Y Coordinate

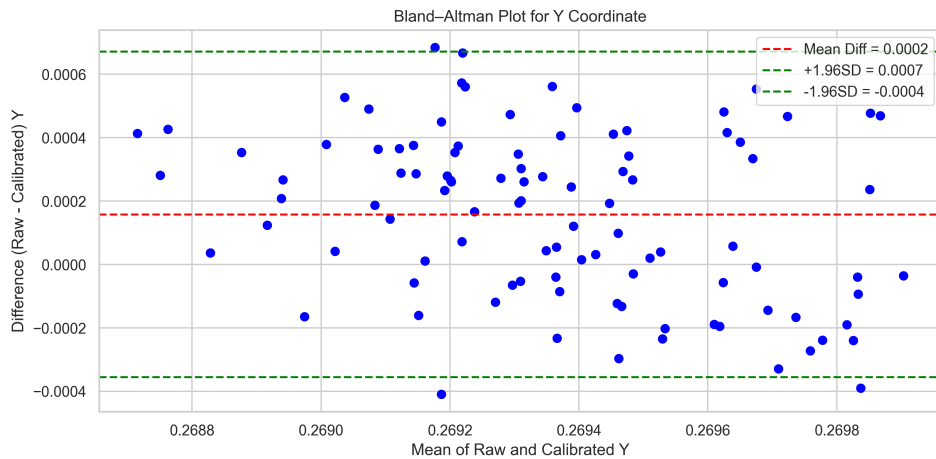


Figure 3.3: Bland–Altman Plot for Y Coordinate.

3.4.1. Overview

- **X-axis:** Mean of raw and calibrated Y coordinates for each landmark.

- **Y-axis:** Difference between raw and calibrated Y coordinates (Raw – Calibrated).
- **Key Elements:**
 - **Red Dashed Line:** Mean difference (0.0002), representing a slight positive bias.
 - **Green Dashed Lines:** Limits of agreement (± 1.96 SD), ranging from -0.0004 to 0.0007 .

3.4.2. Interpretation

1. **Mean Difference:** The slight positive bias (0.0002) indicates a small upward adjustment in vertical positioning.
2. **Limits of Agreement:** With narrow limits (-0.0004 to 0.0007), vertical corrections are consistent.
3. **Scatter Distribution:** Blue dots are evenly spread about the mean difference line; larger differences occur near the image edges.
4. **Clinical Relevance:** Accurate vertical correction is essential for robust face landmark detection, supporting clinical assessments.

3.5. Overall Insights Across All Bland–Altman Plots

1. Calibration introduces minimal bias across all axes:
 - X-axis: Negligible bias (-0.0001).
 - Y-axis: Slight positive bias (0.0002).
 - Z-axis: Zero bias (0.0000).
2. The narrow limits of agreement confirm high precision in distortion correction:
 - X-axis: $[-0.0013, 0.0011]$.
 - Y-axis: $[-0.0004, 0.0007]$.
 - Z-axis: $[-0.0007, 0.0008]$.

3. The scatter distributions show no significant trends, indicating consistent calibration across varying spatial positions.
4. These results validate that camera calibration effectively corrects lens distortion while preserving spatial accuracy, which is crucial for clinical applications.

3.6. Clinical Significance

The Bland–Altman plots confirm that calibration ensures reliable face landmark detection with minimal errors across all axes. This reliability is critical for clinical scoring systems such as ARAT, UPDRS, and Fugl-Meyer assessments, where precise 3D spatial representation is essential for accurate motion analysis.

4) Box Plots

4.1. Introduction

Box plots provide a statistical summary of the error distributions for the X, Y, and Z coordinates, as well as the Euclidean distance. They highlight the median, interquartile range (IQR), and potential outliers, thereby offering insights into the precision and consistency of the calibration process.

4.2. 1. Box Plot for X Error

4.2.1. Overview

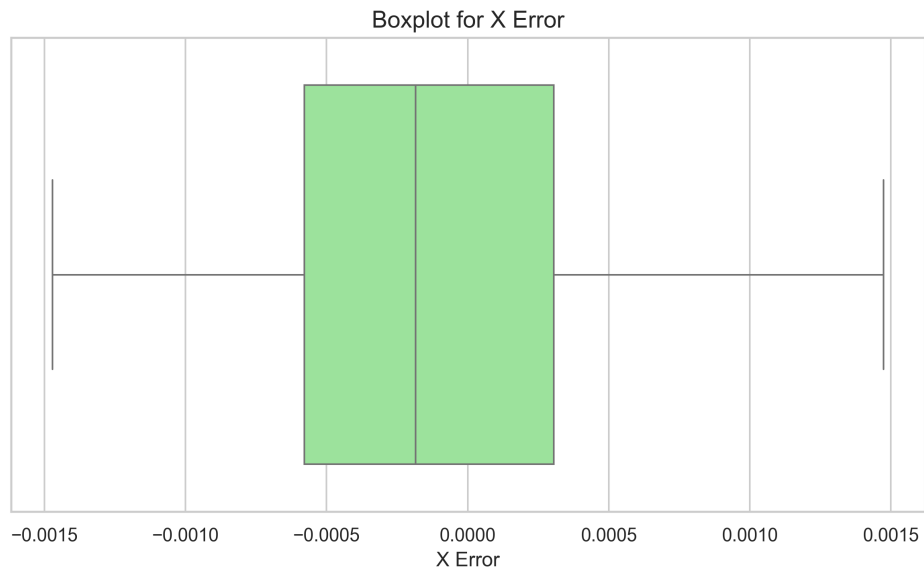


Figure 4.1: Box Plot for X Error.

- **X-axis:** X Error (difference between raw and calibrated X coordinates).
- **Y-axis:** Not applicable (this is a univariate plot).

- **Key Elements:**

- **Green Box:** Represents the interquartile range (IQR), containing 50% of the data.
- **Horizontal Line Inside the Box:** The median of the X error distribution.
- **Whiskers:** Extend to 1.5 times the IQR or the range of non-outlier data.
- **No Outliers:** No points fall beyond the whiskers.

4.2.2. Interpretation

1. **Median and Spread:** The median X error is close to zero, indicating negligible systematic bias. The narrow IQR shows that most errors are small and consistent.
2. **Whiskers:** They extend symmetrically, indicating minimal extreme values.
3. **Clinical Relevance:** Small and consistent X errors ensure accurate horizontal positioning of face landmarks.

4.3. 2. Box Plot for Z Error

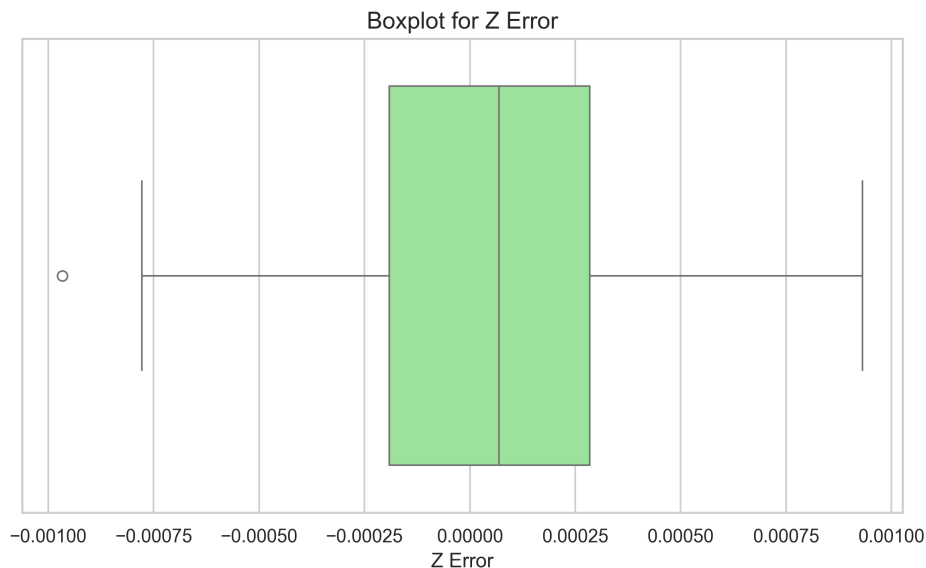


Figure 4.2: Box Plot for Z Error.

4.3.1. Overview

- **X-axis:** Z Error (difference between raw and calibrated Z coordinates).
- **Key Elements:**
 - Structure similar to the X error plot.
 - A single outlier is visible on the left side.

4.3.2. Interpretation

1. **Median and Spread:** The median Z error is zero, confirming no systematic bias in depth corrections. The narrow IQR reflects consistent depth adjustments.
2. **Outlier:** A single outlier indicates an unusually large depth correction for one landmark or frame, possibly due to strong distortion near the edges.
3. **Whiskers:** Extend symmetrically, suggesting balanced error distribution aside from the outlier.
4. **Clinical Relevance:** Precise depth correction is critical for reliable 3D motion analysis.

4.4. 3. Box Plot for Y Error

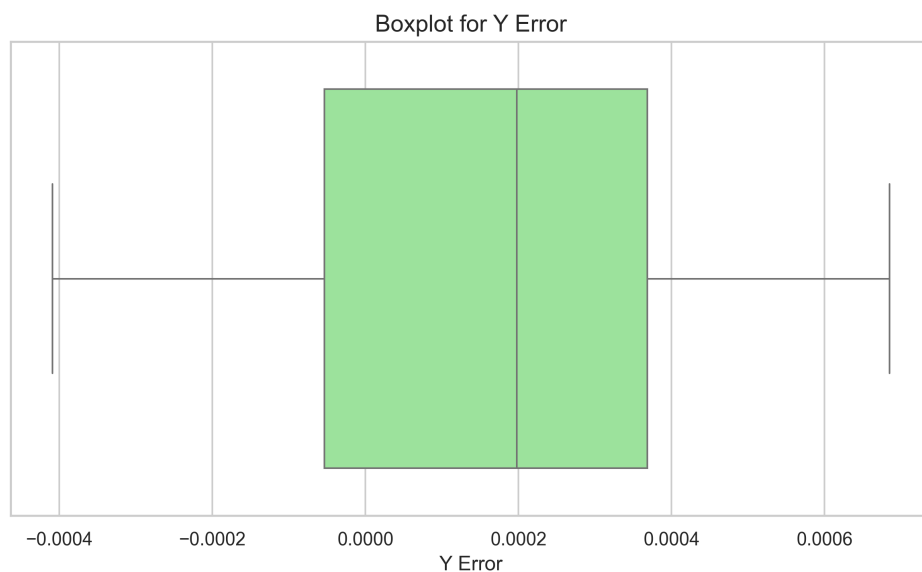


Figure 4.3: Box Plot for Y Error.

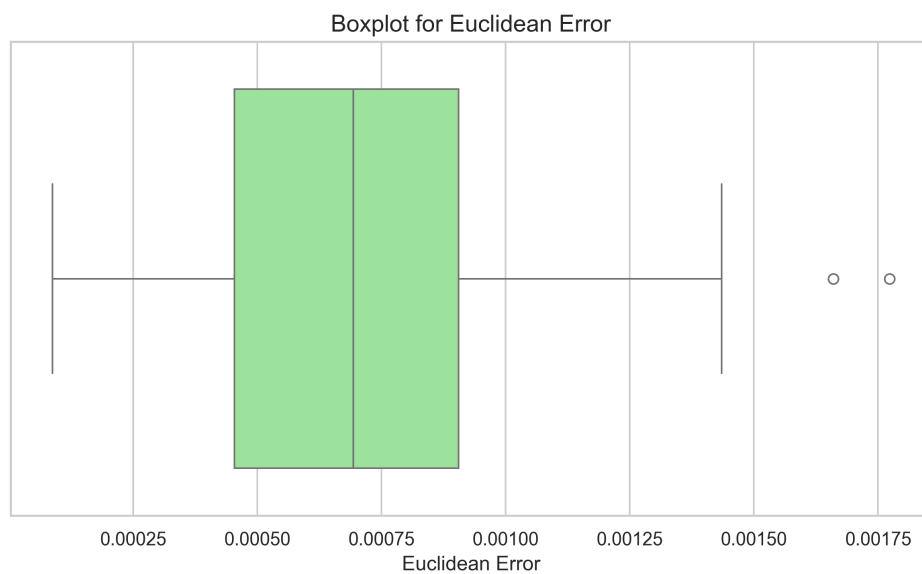
4.4.1. Overview

- **X-axis:** Y Error (difference between raw and calibrated Y coordinates).
- **Key Elements:**
 - Structure is similar to the X error box plot.
 - No visible outliers are observed.

4.4.2. Interpretation

1. **Median and Spread:** The median Y error is slightly above zero, suggesting a small positive bias in vertical corrections. The narrow IQR shows consistent adjustments.
2. **Whiskers:** Extend symmetrically without extreme values, confirming stability in vertical corrections.
3. **Clinical Relevance:** Minimal Y errors ensure accurate vertical positioning, which is vital for motion tracking in clinical assessments.

4.5. 4. Box Plot for Euclidean Error

**Figure 4.4:** Box Plot for Euclidean Error.

4.5.1. Overview

- **X-axis:** Euclidean Error (3D distance between raw and calibrated landmarks).
- **Key Elements:**
 - Similar structure as the other box plots.
 - Two outliers are visible on the right side.

4.5.2. Interpretation

1. **Median and Spread:** The median Euclidean error is small (approximately 0.0005), indicating high overall calibration accuracy. The IQR is slightly wider than individual axis errors, reflecting cumulative variability.
2. **Outliers:** Two outliers denote frames or landmarks with unusually large calibration shifts, possibly due to extreme distortion effects or calibration anomalies.
3. **Whiskers:** Extend symmetrically, with most errors falling within a narrow range, showing consistent calibration performance.
4. **Clinical Relevance:** The small Euclidean errors ensure precise 3D spatial representation, which is essential for clinical scoring tasks.

4.6. Overall Insights Across All Box Plots

1. Calibration introduces minimal errors across all dimensions:
 - Median errors are close to zero.
 - Narrow IQRs indicate consistent corrections.
2. Outliers are rare and likely associated with extreme distortion effects near the image edges or specific calibration anomalies.
3. These results validate that camera calibration effectively reduces distortion while preserving spatial accuracy for face landmark detection.

5) Cumulative Distribution Function (CDF) Plots

5.1. Introduction

Cumulative Distribution Function (CDF) plots provide a statistical summary of the calibration error distributions. They display the cumulative fraction of data points below a given error value, offering valuable insights into the overall error distribution and calibration accuracy for each spatial dimension (X, Y, Z) as well as the overall Euclidean error.

5.2. 1. CDF for Y Error

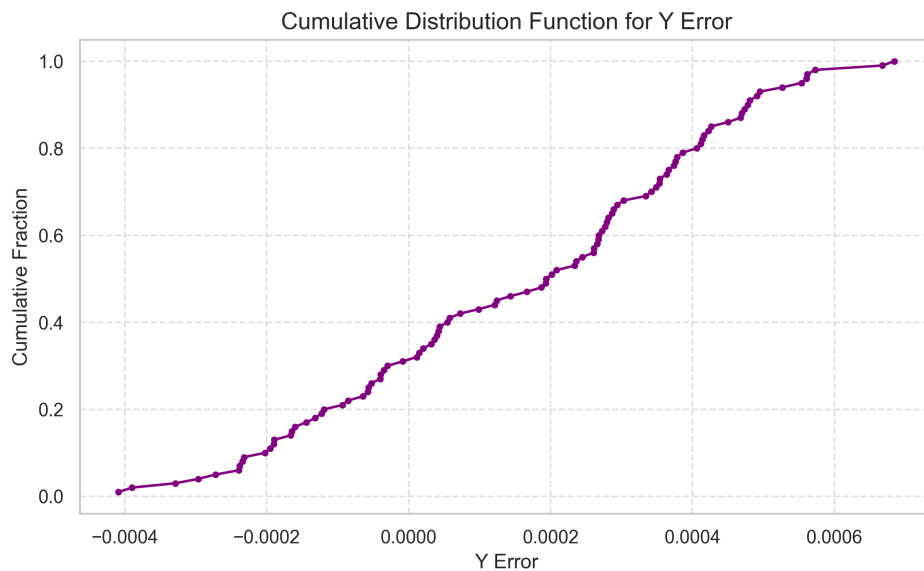


Figure 5.1: CDF Plot for Y Error.

5.2.1. Overview

- **X-axis:** Y Error (difference between raw and calibrated Y coordinates).
- **Y-axis:** Cumulative fraction of data points, starting at 0 and approaching 1.

5.2.2. Interpretation

1. **Symmetry Around Zero:** The curve is centered around zero, indicating that most Y errors are small and evenly distributed between positive and negative values—suggesting minimal bias in vertical corrections.
2. **Steep Slope:** A steep slope near zero shows that a large proportion of errors are very close to zero, reflecting high precision in Y coordinate calibration.
3. **Range:** Y error values typically range from approximately -0.0004 to 0.0006, with the majority of errors concentrated near zero.
4. **Clinical Relevance:** Accurate vertical corrections ensure reliable face landmark detection, critical for clinical scoring systems like ARAT and UPDRS.

5.3. 2. CDF for X Error

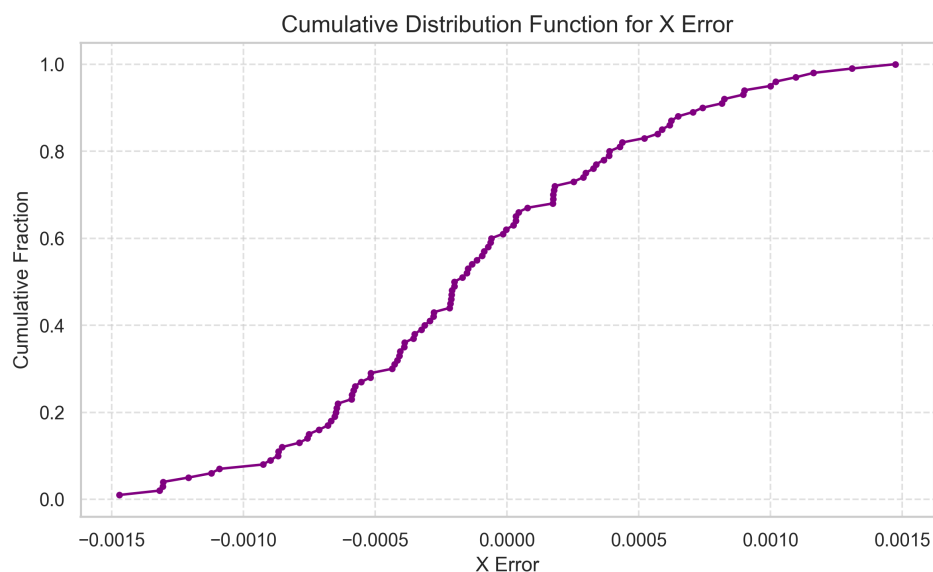


Figure 5.2: CDF Plot for X Error.

5.3.1. Overview

- **X-axis:** X Error (difference between raw and calibrated X coordinates).
- **Y-axis:** Cumulative fraction of data points.

5.3.2. Interpretation

1. **Symmetry Around Zero:** The curve is centered around zero, confirming negligible bias in horizontal corrections.
2. **Steep Slope Near Zero:** Indicates that most X errors are very small, reflecting consistent calibration accuracy for horizontal positioning.
3. **Range:** X error values typically range from approximately -0.0015 to 0.0015, with the majority of errors concentrated near zero.
4. **Clinical Relevance:** Minimal X errors ensure accurate horizontal positioning of face landmarks, essential for reliable motion analysis in clinical assessments.

5.4. 3. CDF for Euclidean Error

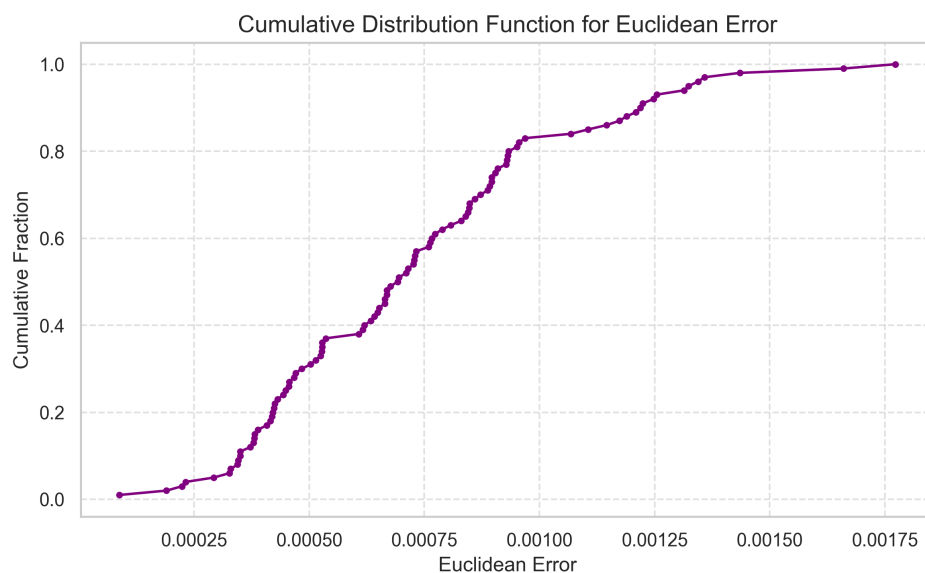


Figure 5.3: CDF Plot for Euclidean Error.

5.4.1. Overview

- **X-axis:** Euclidean Error (3D distance between raw and calibrated landmarks).
- **Y-axis:** Cumulative fraction of data points.

5.4.2. Interpretation

1. **Positive Values Only:** As the Euclidean error is inherently positive, the curve starts from the minimum error (approximately 0.00025) and extends up to the maximum (around 0.00175).
2. **Steep Initial Slope:** A steep slope at the lower end indicates that most Euclidean errors are small, confirming high overall calibration accuracy across all dimensions.
3. **Range:** The majority of Euclidean errors are concentrated below 0.001, with only a small fraction extending up to 0.00175.
4. **Clinical Relevance:** Small Euclidean errors ensure precise 3D spatial representation of face landmarks, which is crucial for clinical scoring tasks, such as in Fugl-Meyer assessments.

5.5. 4. CDF for Z Error

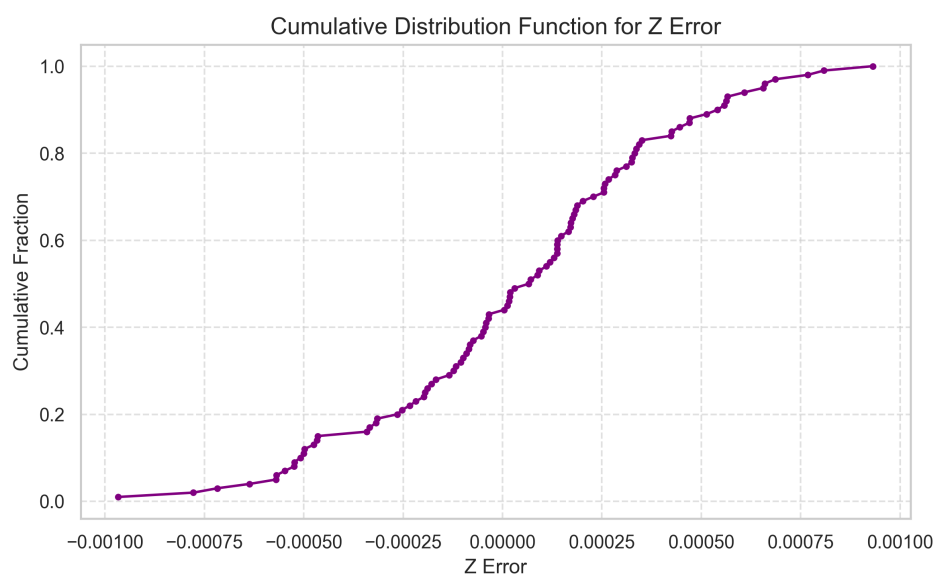


Figure 5.4: CDF Plot for Z Error.

5.5.1. Overview

- **X-axis:** Z Error (difference between raw and calibrated Z coordinates).
- **Y-axis:** Cumulative fraction of data points.

5.5.2. Interpretation

1. **Symmetry Around Zero:** The curve is centered around zero, confirming no significant bias in depth corrections.
2. **Steep Slope Near Zero:** Indicates that a large proportion of Z errors are close to zero, reflecting consistent depth correction accuracy.
3. **Range:** Z error values typically range from approximately -0.001 to 0.001, with the majority of values concentrated near zero.
4. **Clinical Relevance:** Accurate depth corrections are critical for reliable 3D motion analysis in clinical assessments, such as UPDRS and ARAT.

5.6. Overall Insights Across All CDF Plots

1. Calibration introduces minimal errors across all dimensions (X, Y, Z):
 - The curves for X, Y, and Z errors are symmetrically centered around zero.
 - Steep slopes near zero indicate that most errors are very small, reflecting high precision.
 - The narrow ranges confirm consistent calibration performance.
2. Most Euclidean errors are concentrated below 0.0015, validating the overall accuracy of the calibration process.
3. These results confirm that camera calibration effectively reduces distortion while maintaining spatial accuracy for face landmark detection in clinical applications.

6) Histogram with KDE Plots

6.1. Introduction

Histograms with Kernel Density Estimation (KDE) plots provide a visual summary of the error distributions for Euclidean, X, Y, and Z coordinates. The histogram shows the frequency of error values, while the KDE curve offers a smoothed approximation of the underlying probability distribution, helping assess the spread, bias, and skewness of calibration errors.

6.2. 1. Histogram with KDE for Euclidean Error

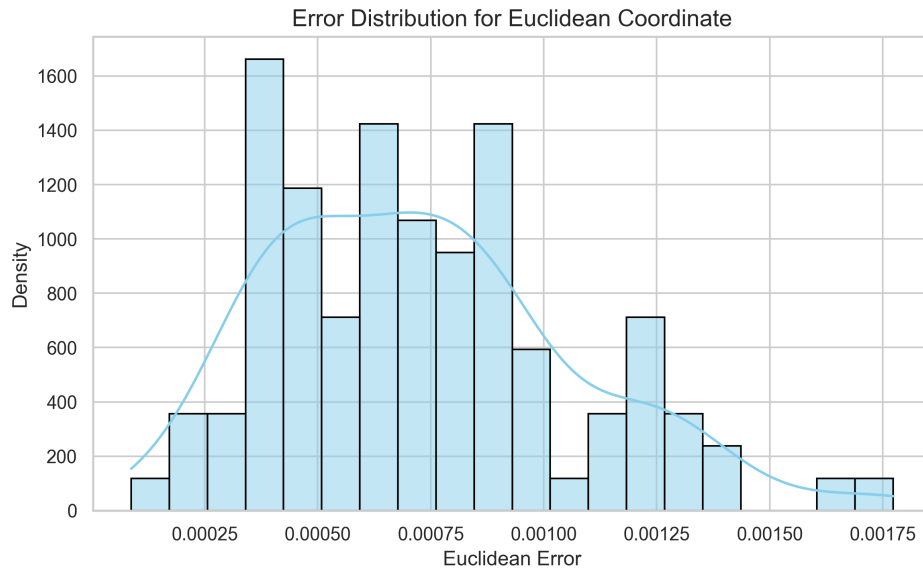


Figure 6.1: Histogram with KDE for Euclidean Error.

6.2.1. Overview

- **X-axis:** Euclidean error (3D distance between raw and calibrated landmarks).

- **Y-axis:** Density (normalized frequency).
- **Key Elements:**
 - Bars represent the histogram of error frequencies.
 - A blue KDE curve overlays the histogram to indicate the smoothed density.

6.2.2. Interpretation

1. **Shape and Range:** The unimodal and slightly right-skewed distribution has most values near 0.0005, ranging from 0.00025 to 0.00175.
2. **Peak:** The peak aligns with the most common error magnitude, confirming low calibration error overall.
3. **Tail:** The longer right tail suggests occasional larger errors, possibly from distorted regions near image edges.
4. **Clinical Relevance:** Precise 3D localization of face landmarks is vital for scoring protocols like ARAT and UPDRS.

6.3. 2. Histogram with KDE for X Error

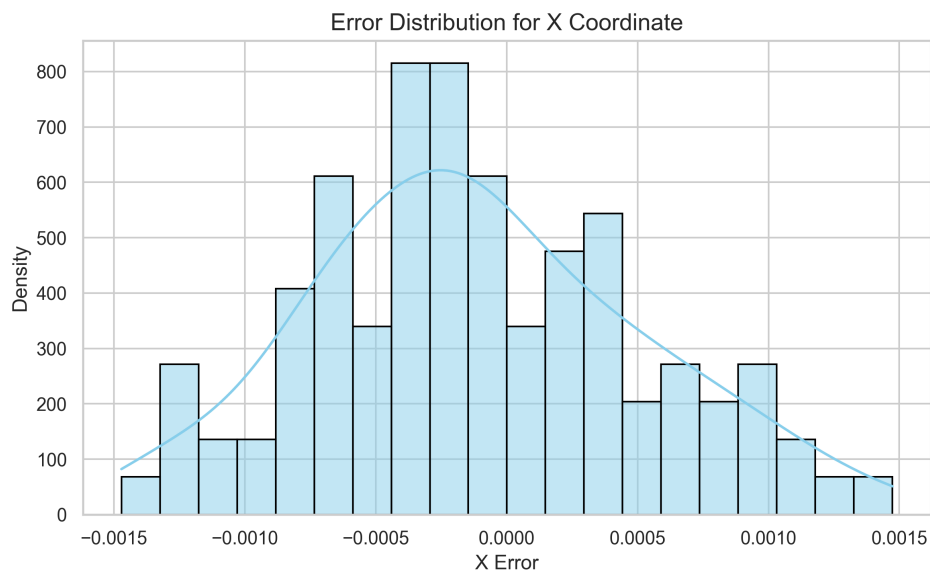


Figure 6.2: Histogram with KDE for X Error.

6.3.1. Overview

- **X-axis:** X error (difference between raw and calibrated X coordinates).
- **Y-axis:** Density.
- Symmetrical histogram with KDE curve centered around zero.

6.3.2. Interpretation

1. **Shape and Centering:** The distribution is symmetric and centered at zero, indicating no horizontal bias.
2. **Spread:** Most errors fall between -0.0005 and 0.0005, with overall range from -0.0015 to 0.0015.
3. **Peak:** The KDE peaks at zero, reinforcing consistent horizontal calibration.
4. **Clinical Relevance:** Reliable horizontal positioning is essential for motion tracking in clinical scoring.

6.4. 3. Histogram with KDE for Y Error

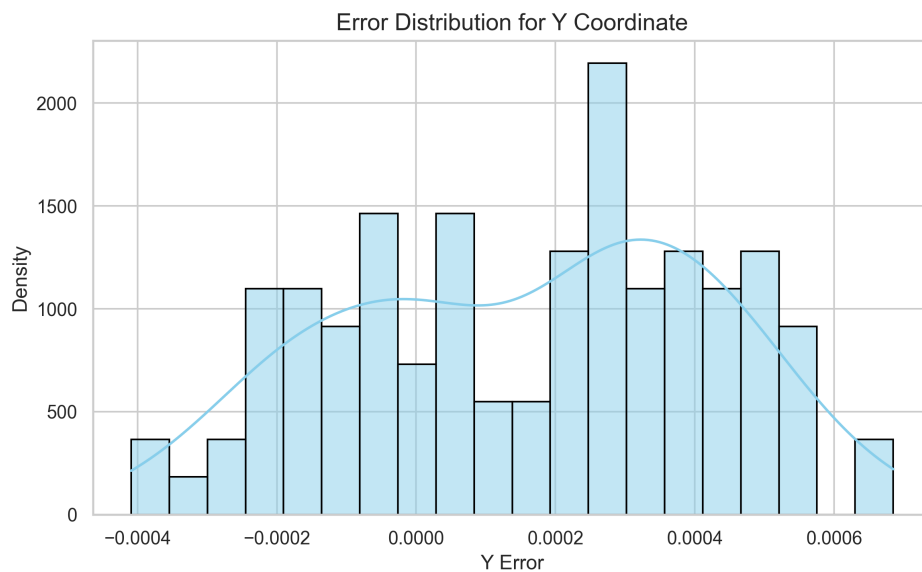


Figure 6.3: Histogram with KDE for Y Error.

6.4.1. Overview

- **X-axis:** Y error (difference between raw and calibrated Y coordinates).
- **Y-axis:** Density.
- KDE shows a slight positive skew, with the peak slightly above zero.

6.4.2. Interpretation

1. **Shape and Bias:** Slight skew to the right indicates a minor positive bias in vertical corrections.
2. **Spread:** The range of Y errors is approximately -0.0004 to 0.0006, with most values near 0.0002.
3. **Peak:** KDE curve peaks just above zero, confirming the vertical calibration bias is minimal.
4. **Clinical Relevance:** Accurate vertical correction ensures consistent facial landmark positioning for evaluations like Fugl-Meyer.

6.5. 4. Histogram with KDE for Z Error

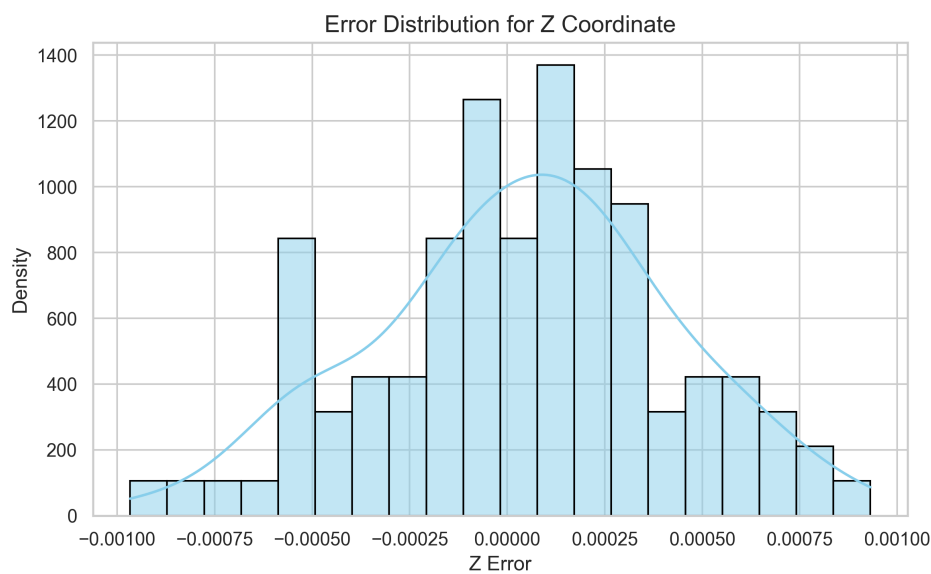


Figure 6.4: Histogram with KDE for Z Error.

6.5.1. Overview

- **X-axis:** Z error (difference between raw and calibrated Z coordinates).
- **Y-axis:** Density.
- Symmetrical histogram with KDE centered around zero.

6.5.2. Interpretation

1. **Shape and Centering:** Distribution is symmetric with peak at zero, indicating unbiased depth correction.
2. **Spread:** Z errors span -0.001 to 0.001, with most errors tightly concentrated around the center.
3. **Peak:** KDE confirms precision in Z correction with minimal spread.
4. **Clinical Relevance:** Reliable depth alignment enhances clinical motion analysis protocols like UPDRS and ARAT.

6.6. Overall Insights Across All Histogram with KDE Plots

1. Calibration introduces minimal errors across all dimensions:
 - Most errors are tightly clustered around zero.
 - Distributions are narrow with slight skew only in Y error.
2. Symmetrical KDEs for X and Z confirm unbiased corrections, while Y shows a small positive shift.
3. The Euclidean error distribution validates overall spatial accuracy.
4. These plots confirm the calibration system's robustness for clinical-grade facial landmark tracking.

7) Scatter Regression Plots

7.1. Introduction

Scatter regression plots visualize the relationship between raw (uncalibrated) and calibrated coordinates for each axis (X, Y, Z). By fitting a linear regression line to the scatter of values, these plots quantify how closely the calibration process aligns corrected coordinates to the original spatial structure.

7.2. 1. Scatter Regression Plot for Y Coordinate

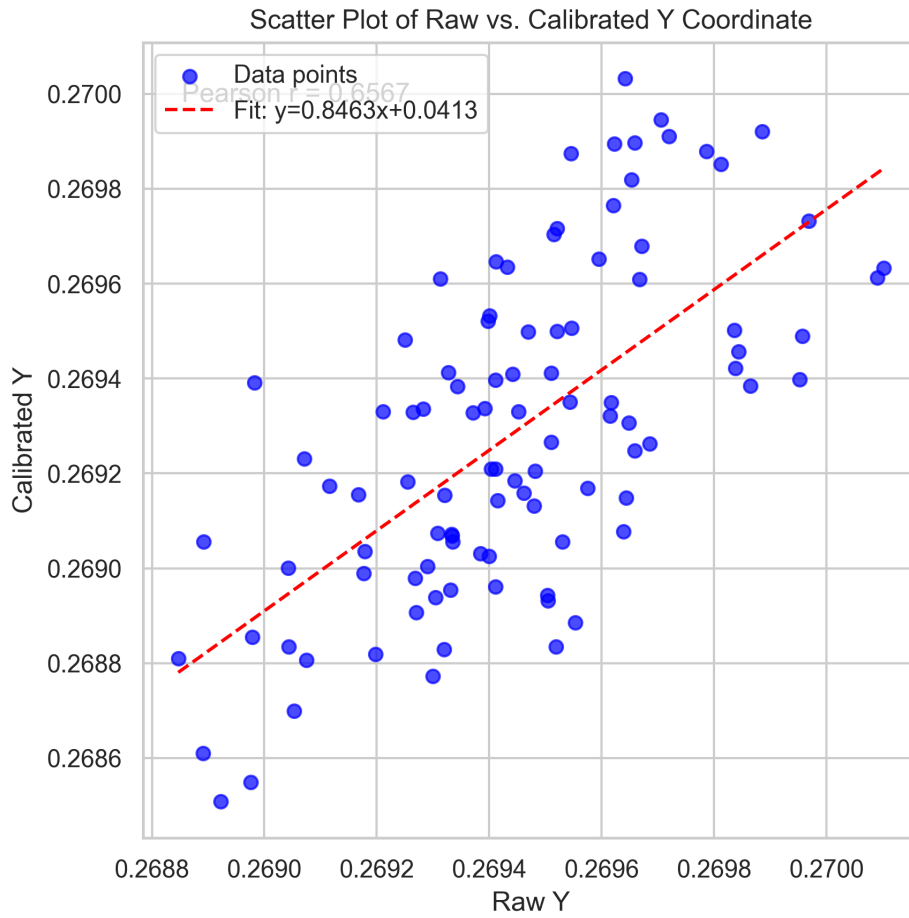


Figure 7.1: Scatter Regression Plot for Y Coordinate.

7.2.1. Overview

- **X-axis:** Raw Y coordinate.
- **Y-axis:** Calibrated Y coordinate.
- **Key Elements:**
 - Blue dots represent individual data points.
 - Red dashed line represents the fitted regression line: $y = 0.8463x + 0.0413$.
 - Pearson correlation coefficient: $R = 0.6567$.

7.2.2. Interpretation

1. **Correlation:** Moderate positive correlation indicates noticeable vertical adjustments due to calibration.
2. **Regression Line:** Slope less than 1 suggests that calibrated Y values tend to be smaller than raw values.
3. **Scatter Distribution:** Moderate spread reflects variability in vertical distortion correction.
4. **Clinical Relevance:** Vertical accuracy is essential for scoring systems such as ARAT and UPDRS.

7.3. 2. Scatter Regression Plot for X Coordinate

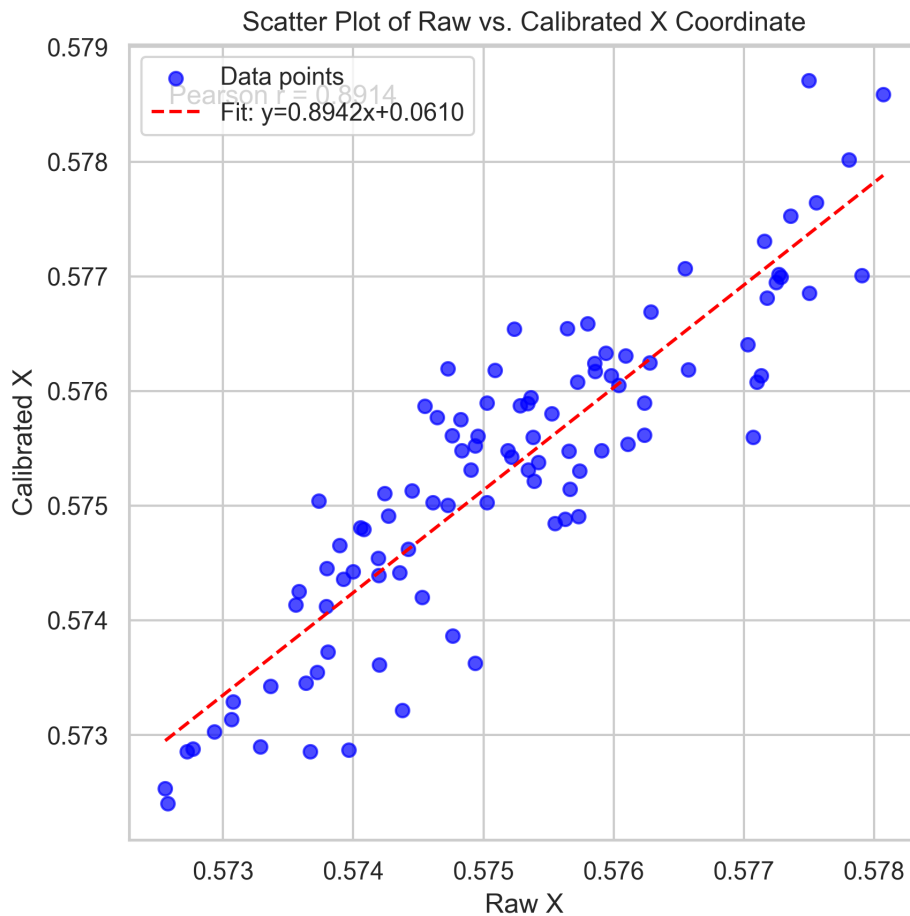


Figure 7.2: Scatter Regression Plot for X Coordinate.

7.3.1. Overview

- **X-axis:** Raw X coordinate.
- **Y-axis:** Calibrated X coordinate.
- **Key Elements:**
 - Blue dots represent raw vs. calibrated values.
 - Red dashed regression line: $y = 0.8942x + 0.0610$.
 - Pearson correlation: $R = 0.8914$.

7.3.2. Interpretation

1. **Correlation:** Strong correlation suggests minimal adjustment in horizontal positioning.
2. **Regression Line:** Slope near 1 reflects alignment between raw and calibrated coordinates.
3. **Scatter Distribution:** Tight clustering around the line indicates stable corrections.
4. **Clinical Relevance:** Horizontal consistency is critical for motion scoring applications like Fugl-Meyer.

7.4. 3. Scatter Regression Plot for Z Coordinate

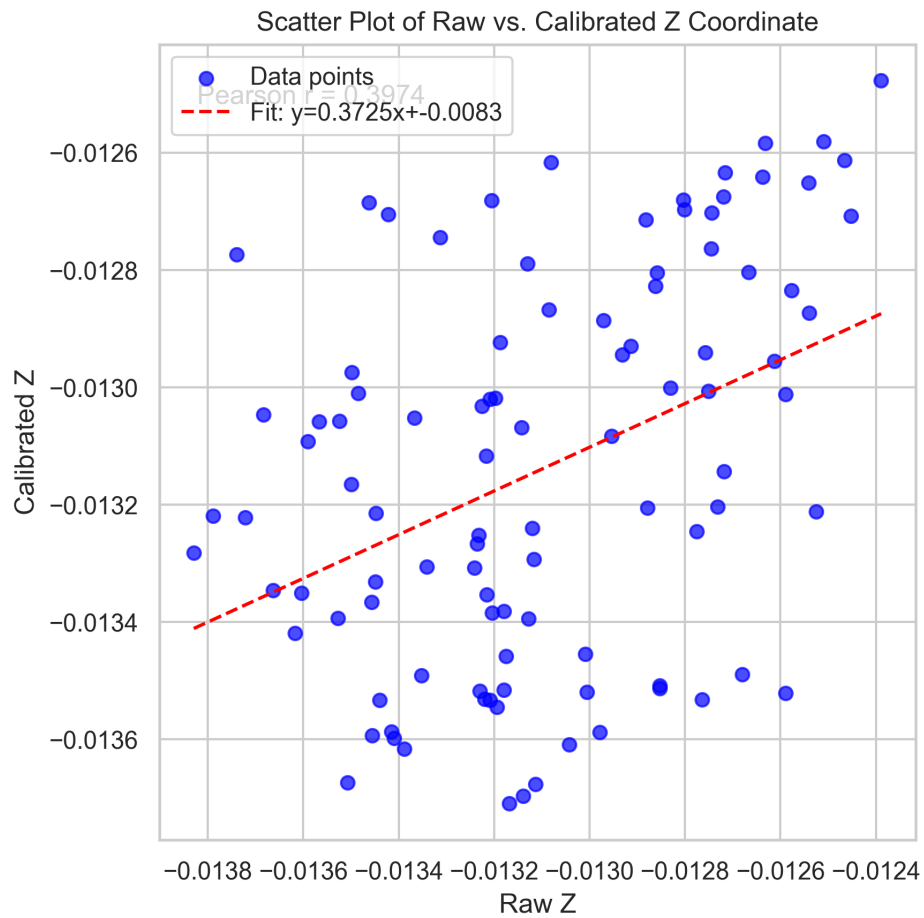


Figure 7.3: Scatter Regression Plot for Z Coordinate.

7.4.1. Overview

- **X-axis:** Raw Z coordinate.
- **Y-axis:** Calibrated Z coordinate.
- **Key Elements:**
 - Blue dots represent raw-calibrated point pairs.
 - Red dashed regression line: $y = 0.3725x - 0.0083$.
 - Pearson correlation: $R = 0.3974$.

7.4.2. Interpretation

1. **Correlation:** Weak correlation indicates substantial variation in depth adjustments.
2. **Regression Line:** A shallow slope reflects significant calibration-driven compression of depth values.
3. **Scatter Distribution:** High variance around the regression line illustrates inconsistency in depth correction.
4. **Clinical Relevance:** Depth accuracy is essential for capturing fine motion, especially in assessments like UPDRS and ARAT.

7.5. Overall Insights Across All Scatter Regression Plots

1. Correlation Summary:

- X axis: $R = 0.8914$ (strong correlation).
- Y axis: $R = 0.6567$ (moderate correlation).
- Z axis: $R = 0.3974$ (weak correlation).

2. Regression Slopes:

- X: ≈ 0.89 , Y: ≈ 0.85 , Z: ≈ 0.37 .
- Lower slopes suggest greater calibration correction.

3. Scatter Spread:

- Tightest for X, moderate for Y, widest for Z—reflecting the typical distribution of lens distortion effects.

4. Clinical Relevance:

- Ensures precise tracking for clinical applications by correcting spatial inaccuracies across all axes.